Supplementary Material

Autonomous Vision-Architecture Research with a Language-Model Agent: A Longitudinal Case Study

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This document accompanies the main paper. It records the implementation stack and the human-authored instruction documents that bounded the agent's behaviour (Appendix A), reproduces the complete per-hypothesis record in a single tabular projection (Appendix B), and collects the additional tables and figures omitted from the main paper for space (Appendix C). The remaining supplementary artefacts (the research brief and workflow template, the full research log, the per-hypothesis CSV, the figure-generation scripts, the per-phase champion configurations, and the Slurm submission templates) are provided as separate files in the accompanying archive and are indexed by its `README`.

A Implementation Details

The main text described the loop at the level of research affordances rather than specific tools. This appendix records the stack used in the case study and the human-authored documents that bounded the agent's behaviour. We do not claim that any single component is essential; we list them so that other groups can either replicate or substitute. These supplementary materials (the research log, the per-hypothesis CSV used to generate every figure in the paper, and the figure-generation scripts) are provided alongside the submission; the training code and tracking-service exports will be released under a permanent DOI upon publication.

A.1 Agent and Tooling

The case-study run used Anthropic's Claude (Opus 4 series) accessed through the Claude Code command-line interface, so that all tool use is achieved through ordinary shell and file-system primitives (file read, file edit, shell command, search). Any comparable CLI-capable frontier LLM should be a drop-in replacement. The specific model identifier and version tags at the time of each phase are recorded in the accompanying materials. No fine-tuning, RAG layer, or domain-specific scaffold was used; the agent's only persistent memory across sessions is the research log described below.

Tool access was managed through the Model Context Protocol (MCP). Three external services were exposed through MCP from the second phase onward: a preprint search and full-text service for arXiv, a documentation-question-answering service for GitHub repositories, and a model-and-dataset hub. Each is presented to the agent through a uniform tool interface so that literature retrieval, repository inspection, and dataset enumeration follow the same call pattern.

A.2 Compute Infrastructure

Experiments ran on a shared HPC allocation managed by Slurm, with a mix of accelerator generations. Small-scale phases used a single GPU per job; the large-scale phase used two GPUs per job (capped by the per-user allocation limit at our site). The agent was explicitly prohibited from running training on interactive login hosts; this constraint was enforced by an instruction in the research brief and, after one observed violation in early Phase 1, by a wrapper script that rejected training-command invocations outside of a Slurm job context. Aggregate compute is reported in the main paper.

A typical hypothesis cycle consisted of: literature query (a few seconds, optional), code or configuration edit (under a minute of agent wall-clock), Slurm submission (seconds), queue wait (highly variable, often the dominant component on shared infrastructure), training (minutes to days depending on phase), and result interpretation and log entry (seconds of agent time). The end-of-turn-after-submission discipline (described in the main paper) ensured that the agent did not consume tokens while the job was queued or training.

A.3 Source, Experiment, and Memory Management

A single Git repository, with one branch per phase and one commit per accepted hypothesis. Rejected hypotheses were reverted from the working tree but remain reconstructible from the research log. Commit messages followed a fixed template (`phase: short title (hypothesis ID)`), giving a one-line audit trail directly inside source control.

Experiment tracking used Weights & Biases, with one project per phase. Every Slurm job logged per-epoch training loss, validation top-1 and top-5, parameter count, FLOPs, and a fixed set of system statistics. The agent was expected to reference training curves (not only final metrics) when interpreting ambiguous results; we observed it doing so on several occasions where a final number was within noise but a divergent training curve disqualified the run.

Persistent memory took the form of a single Markdown research log containing one appended entry per hypothesis in the fixed template described in Section A.6. The agent was required to read this file in full at the start of every session (after an extended pause or context reset). On long programmes the log eventually exceeded the agent's effective attention window, at which point a compact agent-authored *empirical-laws synthesis* document was introduced (regenerated periodically as a forced exercise in summarisation) and read once per session in addition to the log.

A.4 Persistence and Control Hand-off

The agent process was kept alive across user disconnects in a persistent terminal session. To return control to the agent on job completion without requiring the agent to poll the scheduler (which would have consumed tokens and triggered the failure mode of mis-classifying queued jobs as running), a short shell utility watched for the appearance of the per-job log file's termination marker and re-prompted the agent in the same terminal session. We treat this as an implementation detail rather than a methodological contribution; any equivalent control-hand-off mechanism (a daemon, a cron entry, a job-completion hook) would serve the same purpose.

A.5 Human-Authored Instruction Documents

Three human-authored documents bounded the agent's behaviour. Their exact wording (including revisions over the programme) is included in the supplementary materials. We here summarise their structure.

The research brief. The brief is two-layered: a fuller research-programme document (`RESEARCH_PROGRAM.md` in the supplementary) and a shorter agent-facing brief (`RESEARCH_BRIEF.md`) that points to it. Together they specify: the scientific question (here, designing a channel-primary vision transformer under a pure channel-attention constraint); the design constraints (no spatial self-attention at any stage, parameter and FLOP budget per phase, dataset for each phase); the success criterion (validation top-1 on a fixed held-out set); and a list of *illustrative* directions intended to seed the agent's literature exploration without prescribing the answer (see Section A.7 below). The brief was rewritten between phases to reflect transitions (recorded among the human interventions in the main paper); the rewritten version was always presented as the agent's authoritative single source of truth, and the previous version was archived.

The workflow template. A separate document specifying the per-hypothesis loop (described in the main paper), the expected structure of each log entry, and explicit prohibitions: no scheduler polling, no training on the interactive login host, one experiment in flight at a time, no architectural changes without a corresponding source-control commit, and a budget check (parameter count and estimated FLOPs) before any submission. This document was updated only to add reliability constraints in response to observed autonomy failures (Section A.8); the scientific content lived in the research brief.

The literature-grounding requirement. A short rule, added at the Phase 1/1b boundary: every hypothesis must either cite a concrete prior result (a paper, repository, or model card) or explicitly declare itself an internal extrapolation. This rule is short but had a visible effect on the agent's behaviour: it shifted hypothesis selection from configuration tweaks toward code-level mechanism changes. We attribute the effect to the rule forcing the agent to ground its proposal in a specific external reference rather than in pattern completion over its recent context.

A.6 Per-Hypothesis Log Template

Each entry in the research log follows a fixed seven-field template:

1. **Identifier and title.** A short symbolic identifier (`H1`, `P2-H7`, `H3-12`) and a one-line title naming the single variable changed.
2. **Motivation.** Two or three sentences stating the hypothesis to be tested and why it is worth testing now.
3. **Literature basis.** A specific citation (paper, repository, or model card) or an explicit declaration that the hypothesis is an internal extrapolation. The literature-grounding rule (Section A.5) requires this field to be non-empty from Phase 1b onward.
4. **Change description.** The exact diff applied: which configuration values, which code paths. This field is the source of the CFG/COD change-type classification used throughout the paper.
5. **Budget.** Parameter count, FLOP estimate, and expected wall-clock; all three are recomputed by a budget-check utility before submission.
6. **Result.** Final validation metric, Δ vs. current champion, and a short interpretive sentence written by the agent after the run.
7. **Status.** One of `pushed`, `discarded`, `pending`, with the corresponding Git SHA if pushed.

The full log is provided as a Markdown file in the supplementary materials. A tabular projection of all hypotheses appears in Appendix B.

A.7 Illustrative Directions in the Phase 3 Brief

The Phase 3 brief included a non-exhaustive list of five illustrative structural directions, intended to seed (not constrain) the agent's hypothesis generation: funnelling architectures with extra extreme-spatial-compression stages, factorised or low-rank channel attention, sparse top-k channel attention, hierarchical local-plus-global channel attention, and linear or state-space channel attention. The brief explicitly framed this as a *menu*, instructed the agent to synthesise novel ideas from the literature, and required a per-hypothesis self-audit of which novelty category the proposal fell into. We treat the menu as a documented direction-shaping intervention (recorded among the human interventions in the main paper), because the agent's Phase 3 hypothesis set ended up including adaptations of all five.

A.8 Autonomy Failure Modes Encountered

Several classes of autonomy failure appeared during the programme and were each fixed once. We list them because they are likely to recur in any similar setup.

1. **Premature turn termination.** In early Phase 1 the agent would sometimes declare a turn finished after merely *submitting* a Slurm job, without waiting for or interpreting the result. Fixed by an explicit "end your turn after submission and wait for an external signal" instruction (see the main paper) and the control-hand-off mechanism of Section A.4.
2. **Scheduler mis-reading.** The agent sometimes classified queued jobs as running, leading to spurious result interpretations. Fixed by routing queue-status queries through a small wrapper that returned only `queued` / `running` / `finished`, eliminating the noisy free-text scheduler output.
3. **Login-host training.** Twice in early Phase 1 the agent attempted to start a training process on the interactive login host. Fixed by a wrapper script that rejected training-command invocations outside of a Slurm job context.
4. **Distributed-data-parallel unused-parameter crash.** The agent's H3-18 hypothesis (a Synthesizer-style operator that bypasses the key projection) crashed at iteration 0 because DDP detected an unused parameter. The agent diagnosed the unused projection itself, applied a one-line fix that preserves the parameter's gradient flow without changing the numerics, and resubmitted, without external intervention.
5. **Context-window saturation.** On the longest runs (multi-month elapsed time, hundreds of log entries) the research log eventually exceeded the agent's effective attention window. Fixed by introducing the compact agent-authored synthesis document described in Section A.3.
6. **Token-rate-limit interruptions.** Long sessions occasionally hit the LLM provider's token-rate limit. Fixed by aggressively pruning what the agent was asked to re-read each turn (notes on this are in the research brief), and by tolerating short stalls without retrying through the loop.

Each failure is an instance of the same general pattern: a discipline that is implicit in a human researcher's habits (do not call a job "done" before its results are read, do not run training on a login node, summarise prior work rather than re-read it from scratch each turn) needed to be made explicit in the agent's instructions or enforced mechanically by the loop. We expect any future autonomous-research system to need analogous discipline-encoding work; the specific list above is intended as a starting checklist.

A.9 Approximate Costs

Wall-clock: approximately ten weeks across the three phases (as summarised in the main paper), of which roughly 90% was either queue wait or training and the remaining 10% was agent activity. GPU-hours: of order 2.4 k aggregated across all phases, dominated by the large-scale ImageNet validation runs; the median substantive run was short ($\approx$ 2.5 h). LLM-token consumption: of order tens of millions of tokens across all phases (we did not log this systematically; the figure is a budget-based estimate). Human-researcher involvement: bounded to the interventions listed in the main paper, plus approximately one to two hours per day of passive oversight during active phases (monitoring for autonomy failures and unblocking when they occurred).

We make no claim that this loop is cheaper than a human researcher; our claim is that the *research work* was executed by the agent, with the human role confined to the recorded interventions, and that the resulting behavioural record is worth studying on its own.

A.10 Code and Data Availability

The supplementary materials accompanying this submission contain:

1. the two-layered research brief (`RESEARCH_PROGRAM.md` and `RESEARCH_BRIEF.md`), the workflow template (`AGENT_INSTRUCTIONS.md`), and the tooling-configuration document (`TOOLING.md`), all in their final form;
2. the research log (Markdown), with one entry per hypothesis in the template of Section A.6;
3. the agent-authored empirical-laws synthesis document referenced in Section A.3, and the agent-authored architecture-evolution summary referenced in the main paper;
4. the per-hypothesis CSV used to generate every figure and the longtable in Appendix B;
5. the figure-generation scripts (Python, < 500 lines total);
6. the per-phase Slurm submission script templates and the control-hand-off watcher script (with site-specific identifiers stripped);
7. the architectural specification of the champion at the end of each phase, as YAML.

The full training code (the model implementation, training driver, and per-run configurations of failed hypotheses) is not included in this submission package and will be released under a permanent identifier upon publication. Re-running the architecture on the released training code reproduces the champion accuracy at each phase to within run-to-run variance; we did not budget for multi-seed replications of every accepted hypothesis, and this is one of the limitations discussed in the main paper.

B Per-Hypothesis Record

Table 1 reproduces the full per-hypothesis record in a single tabular projection: one row per submitted hypothesis (and one row per dataset baseline), across all four phase markers (Phase 1, Phase 1b, Phase 2, Phase 3). The table is sorted chronologically within each phase. Identifiers prefixed `H` are CIFAR-10 hypotheses (Phase 1 and Phase 1b), identifiers prefixed `P2-H` are CIFAR-100 hypotheses (Phase 2), and identifiers prefixed `H3-` are ImageNet-1K hypotheses (Phase 3). The *description* column is the agent's own one-line summary of the change, truncated to a fixed width for compactness; the full description, motivation, literature basis, change diff, budget, and outcome interpretation are in the supplementary research log (Section A.10).

Table 1: Full per-hypothesis record extracted from the research log. *Description* is truncated to the first 58 characters of the agent's title; the full motivation, literature citation, change description, and outcome interpretation are in the supplementary log. *Type* distinguishes hypotheses whose only substantive change was a configuration value (CFG) from those that introduced or modified code (COD); all Phase 3 hypotheses were structural and are labelled COD. *Acc* is the validation top-1 (%) at the dataset and protocol of that phase. Δ is the change vs. the running champion at the time the hypothesis was submitted, with the dataset baseline treated as the initial champion for each chain.

ID	Ph.	Description	Type	Acc	Δ	Status
Baseline-C10	1	Dataset baseline	–	69.67	–	
H1	1	Add CA MLP (CA_MLP_RATIO = [4.0, 4.0, 1.0]	cfg	91.08	+21.41	pushed
H2	1	Stage-2-only CA MLP (CA_MLP_RATIO = [0, 0, 1.0]	cfg	89.97	-1.11	discarded
H3	1	Multi-head channel attention (NUM_HEADS = [1, 2, 4]	cfg	89.95	-1.13	discarded
H4	1	Deeper stage-2 backbone (DEPTH = [1, 2, 11]	cfg	91.13	+0.05	pushed
H5	1	Larger early-stage CA MLP (CA_MLP_RATIO = [8.0, 8.0, 1.…	cfg	91.71	+0.58	pushed
H6	1	DW shortcut backbone (DW_SHORTCUT_BACKBONE = true	cfg	92.52	+0.81	pushed
H7	1	Remove stochastic depth (DROP_PATH_RATE = [0,0,0]	cfg	92.98	+0.46	pushed
H8	1	Shorter warmup (WARMUP_EPOCHS = 5	cfg	93.00	+0.02	pushed
H9	1	Higher learning rate (BASE_LR = 1e-3	cfg	93.77	+0.77	pushed
H10	1	Reduce weight decay (WEIGHT_DECAY = 0.01	cfg	93.36	-0.41	discarded
H11	1	Push LR further (BASE_LR = 2e-3	cfg	93.96	+0.19	pushed
H12	1	Push LR to 4e-3 (BASE_LR = 4e-3	cfg	93.41	-0.55	discarded
H13	1	4-stage architecture (NUM_STAGES=4, DIM_EMBED=[64,128,2…	cfg	94.11	+0.15	pushed
H14	1	4-stage with stage-3 stride=1 (14×14 spatial, PATCH_STRI…	cfg	93.98	-0.13	discarded
H15	1	Reallocate depth: 1 stage-1 block → 1 stage-3 block (DEPT…	cfg	93.79	-0.32	discarded
H16	1	Re-tune LR for 4-stage model (BASE_LR = 3e-3	cfg	93.84	-0.27	discarded
H17	1	Larger QKV kernel in stages 0-2 (KERNEL_QKV=[5,5,5,3]	cfg	93.39	-0.72	discarded
H18	1	dw_sep patch embedding at all 4 stages (PATCH_EMBED_ME…	cfg	93.47	-0.64	discarded
H19	1	Deeper stage 0 (DEPTH=[2,2,3,7]	cfg	93.55	-0.56	discarded
H20	1	DEPTH=[1,2,3,8] + CA_MLP_RATIO=[8,8,3.3,1] (stage-3 dep…	cfg	93.68	-0.43	discarded
H21	1	Disable Mixup+CutMix (AUG.MIXUP=0, AUG.CUTMIX=0	cfg	94.79	+0.68	pushed
H22	1	Disable all DeiT augmentation (no RandAugment, no RandomE…	cfg	94.24	-0.55	discarded

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ID	Ph.	Description	Type	Acc	Δ	Status
H23	1	Disable label smoothing (MODEL.LABEL_SMOOTHING=0.0	cfg	94.92	+0.13	pushed
H24	1	Reduce RandomErase probability (REPROB=0.1	cfg	94.71	-0.21	discarded
H25	1	Mild drop path in deepest stage (DROP_PATH_RATE=[0,0,0,…	cfg	94.96	+0.04	pushed
H26	1	Redistribute depth	cfg	94.84	-0.12	discarded
H27	1	Longer warmup (WARMUP_EPOCHS=10	cfg	95.06	+0.10	pushed
H28	1	Even longer warmup (WARMUP_EPOCHS=15	cfg	94.99	-0.07	discarded
H29	1	Progressive drop path [0, 0.01, 0.02, 0.05]	cfg	95.09	+0.03	pushed
H30	1	Layer Scale (CaiT-style, init=1e-4	cod	95.10	+0.01	pushed
H31	1	Attention dropout in stage 3 (ATTN_DROP_RATE=[0,0,0,0.1]	cfg	94.98	-0.12	discarded
H32	1	DW shortcut placed between attention and MLP (DW_SHORTCU…	cfg	95.09	-0.01	discarded
H33	1	SiLU (Swish) activation instead of GELU	cfg	94.91	-0.19	discarded
H34	1	Higher weight decay (WEIGHT_DECAY=0.1	cfg	94.99	-0.11	discarded
H35	1	Lighter RandAugment (rand-m7	cfg	95.03	-0.07	discarded
H36	1	More depth at stage 0 (DEPTH=[2,2,3,7]	cfg	94.79	-0.31	discarded
H37	1	Light Mixup (MIXUP=0.2	cfg	94.83	-0.27	discarded
H38	1	All-dw_sep patch embedding (['dw_sep','dw_sep','dw_se…	cfg	94.80	-0.30	discarded
H39	1	Deep pure CA	cfg	94.03	-1.07	discarded
H40	1	AdamW β_2 =0.98	cfg	94.74	-0.36	discarded
H41	1	Layer Scale init=1e-5	cfg	94.98	-0.12	discarded
H42	1	QK-Normalize (L2-normalize Q and K before dot product	cod	94.22	-0.88	discarded
H43	1b	Learnable Channel Pair Bias (CPB	cod	94.97	-0.13	discarded
H46	1b	Auxiliary deep supervision at stage 2 (intermediate class…	cod	95.70	+0.60	pushed
H47	1b	Per-stage learned attention temperature (log-parameterize…	cod	95.82	+0.12	pushed
H48	1b	Gated attention output (scalar sigmoid gate, channel mode	cod	95.75	-0.07	discarded
H49	1b	Larger DW shortcuts (5×5 kernels, uniform across all stag…	cfg	95.71	-0.11	discarded
H50	1b	Dual auxiliary deep supervision at stages 1 and 2	cod	95.85	+0.03	pushed
H51	1b	Convolutional Position Encoding (CPE) at each stage	cod	95.95	+0.10	pushed
H52	1b	Larger CPE kernel (5×5	cod	96.11	+0.16	pushed
H53	1b	CPE with 7×7 kernel	cod	95.85	-0.26	discarded
H54	1b	SwiGLU MLP (replace GELU with SwiGLU gate in CA MLP blocks	cod	96.23	+0.12	pushed
H55	1b	Per-stage CPE kernel sizes [3, 3, 5, 5]	cod	96.24	+0.01	discarded
H56	1b	DW shortcut kernel verified [3,3,3,3] on H54 base	cfg	96.48	+0.25	pushed
H57	1b	RMSNorm on H56 clean base	cod	96.25	-0.23	discarded
H58	1b	Multi-head channel attention NUM_HEADS [1,2,4,1]	cfg	96.12	-0.36	discarded
H59	1b	Stronger auxiliary loss weights	cod	96.42	-0.06	discarded
H60	1b	Wider stage 2 MLP (ratio 4→6), fewer blocks (3→2	cfg	96.38	-0.10	discarded
H62	1b	Depth redistribution [1,3,3,6]	cfg	96.37	-0.11	discarded
H63	1b	LayerScale 1e-3 (10x larger initial scale	cod	96.20	-0.28	discarded
H64	1b	Triple auxiliary supervision	cod	96.59	+0.11	pushed
H65	1b	Stronger aux weights	cod	96.48	-0.11	discarded
H67	1b	Intra-stage-3 mid-block auxiliary head (after block 4 of 7	cod	96.37	-0.22	discarded
Baseline-C100	2	Dataset baseline	–	81.34	–	

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ID	Ph.	Description	Type	Acc	Δ	Status
P2-H2	2	Stage-3 depth-width tradeoff	cfg	84.11	+2.77	discarded
P2-H3	2	Group channel attention (G=4) at stage 3 + deeper network	cod	81.31	-0.03	discarded
P2-H4	2	Cross-stage channel feature fusion (DuoFormer-inspired	cod	81.72	+0.38	pushed
P2-H5	2	No stochastic depth at 100ep (DROP_PATH=[0,0,0,0]	cfg	81.12	-0.60	discarded
P2-H6	2	Label smoothing LS=0.1 at 100ep	cfg	82.02	+0.30	pushed
P2-H7	2	Mild Mixup=0.4 at 100ep	cfg	82.79	+0.77	pushed
P2-H8	2	Differential channel attention	cod	81.88	-0.91	discarded
P2-H9	2	Intra-stage-3 mid-block auxiliary supervision	cod	82.80	+0.01	pushed
P2-H10	2	Periodic CPE re-injection within stage 3 (interval=4	cod	82.88	+0.08	pushed
P2-H11	2	Larger Q/K/V projection kernel at stage 3 (3×3 → 5×5	cfg	83.16	+0.28	pushed
P2-H12	2	Full-field Q/K/V projection kernel at stage 3 (5×5 → 7×7	cfg	83.11	-0.05	discarded
P2-H13	2	5×5 Q/K/V projection kernel at stage 2 as well	cfg	82.82	-0.34	discarded
P2-H14	2	5×5 DW shortcut kernel at stage 3	cfg	83.17	+0.01	pushed
P2-H15	2	Wider stage-3 embedding (576→640) with fewer depth (12→10	cfg	83.18	+0.01	pushed
P2-H16	2	Cross-stage spatial input fusion (stage-2 features → stag…	cod	82.72	-0.46	discarded
P2-H17	2	Grouped channel attention at stage 3 (G=4 groups of 160 c…	cod	82.98	-0.20	discarded
P2-H18	2	Sigmoid channel attention at stage 3	cod	82.97	-0.21	discarded
P2-H19	2	Value residual for stage-3 channel attention	cod	83.11	-0.07	discarded
P2-H20	2	Dual-scale channel attention (GAP branch	cod	83.16	-0.02	discarded
P2-H21	2	Learned channel-pair attention prior (CHANNEL_PAIR_BIAS	cod	82.89	-0.29	discarded
P2-H22	2	All-stage cross-stage fusion (add stage 0	cod	83.30	+0.12	pushed
P2-H23	2	Second-order statistics pooling in cross-stage fusion	cod	83.30	+0.00	discarded
P2-H24	2	Attentive channel gating (output-side gate on stage-3 cha…	cod	83.13	-0.17	discarded
P2-H25	2	Decoupled V projection	cod	82.82	-0.48	discarded
P2-H26	2	Per-stage periodic CPE re-injection	cod	83.37	+0.07	pushed
P2-H27	2	Stage-3 CPE interval 4→3 (denser spatial refresh in deepe…	cod	82.74	-0.63	discarded
P2-H28	2	5×5 Q/K at stage-2 (extend stage-3 winning kernel size to…	cfg	82.80	-0.57	discarded
P2-H29	2	Register tokens for stage-3 channel attention (K=4	cod	82.70	-0.67	discarded
P2-H30	2	Stage-1 periodic CPE re-injection (interval=1	cod	82.51	-0.86	discarded
P2-H31	2	7×7 CPE kernel	cod	83.03	-0.34	discarded
P2-H32	2	Depth redistribution DEPTH=[1,2,5,10]→[1,2,6,9]	cfg	83.17	-0.20	discarded
Baseline-IN1K	3	Dataset baseline	–	77.65	–	
H3-1	3	Sparse top-k channel attention at stage-3 (k=64 of C=640	cod	77.48	-0.17	discarded
H3-2	3	Dual-resolution channel attention at stage-3 (G=32 super-…	cod	77.50	-0.15	discarded
H3-3	3	Cross-stage channel attention bridge (stage-2 → stage-3	cod	77.55	-0.10	discarded
H3-4	3	Agent attention at stage-3 (factorised low-rank CA via M=…	cod	77.38	-0.27	discarded
H3-5	3	Hierarchical local + global CA at stage-2 (within-group f…	cod	77.45	-0.20	discarded

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ID	Ph.	Description	Type	Acc	Δ	Status
H3-6	3	Funneling tail-block at 3×3 spatial pool (purely additive…	cod	77.59	-0.06	discarded
H3-8	3	Linear-attention funneling tail at full stage-3 spatial (…	cod	62.93	-14.72	discarded
H3-9	3	Linformer-style projected channel attention funnel + shar…	cod	77.79	+0.14	pushed
H3-10	3	Stacked Linformer-style projected channel attention funne…	cod	77.75	-0.04	discarded
H3-11	3	Parallel hybrid funnel	cod	77.58	-0.21	discarded
H3-12	3	Linformer applied to ALL 10 stage-3 in-trunk CA blocks (L…	cod	77.91	+0.12	pushed
H3-13	3	Push Linformer wider	cod	77.65	-0.26	discarded
H3-14	3	Per-block Linformer rank schedule within stage-3	cod	77.67	-0.24	discarded
H3-15	3	Sparse top-k WITHIN the Linformer rank-128 supertoken bas…	cod	77.54	-0.37	discarded
H3-16	3	Gated parallel FULL-RANK CA pathway alongside Linformer a…	cod	77.78	-0.13	discarded
H3-17	3	Sparsemax replacement for softmax inside stage-3 Linforme…	cod	77.74	-0.17	discarded
H3-18	3	Synthesizer-Dense channel attention at stage-3 (REPLACES …	cod	77.39	-0.52	discarded

C Additional Tables and Figures

This appendix reproduces the tables and figures referenced by the main paper but omitted there for space. Each item is presented with the analysis that accompanied it in our extended treatment, so that it can be read on its own rather than as a bare exhibit.

C.1 The three phases of the programme

The programme is organised into three phases separated by human-declared transitions (Table 2). Each transition either expanded the agent's tool surface (literature and code access at the Phase-1/1b boundary) or changed the problem scale (a larger parameter budget and harder dataset at Phase 2, then ImageNet-1K at Phase 3). The agent did not choose these transitions; it operated fully autonomously within each phase, and this phase structure is the only coarse control we imposed on the trajectory. Read down the final column, the end-of-phase acceptance rate falls monotonically (36% → 38% → 32% → 12%) as the search space at each scale is progressively exhausted — the quantitative fingerprint of the saturation dynamics analysed in the main paper.

Table 2: The three research phases of the programme, distinguished by tool access and problem scale. The agent's observed behaviour differs visibly between phases (main paper).

Phase	Hypotheses	Dataset	Scale	Tool surface	Success rate
1	H1–H42	CIFAR-10	5.8M	config/code editing, tracking, source control	36%
1b	H43–H67	CIFAR-10	5.8M	+ literature access (preprints, repositories, hub)	38%
2	P2-H1–P2-H32	CIFAR-100	22M	full toolkit, larger budget	32%
3	H3-1–H3-18	ImageNet-1K	22M	+ multi-GPU distributed configurations	12%

C.2 The complete record of human interventions

Table 3 lists every human action taken across the ten-week programme. Two properties matter for interpreting the behavioural findings. First, the interventions are *bounded*: outside of them the agent proposed, implemented, executed, and interpreted every experiment itself. Second, they are *strategic rather than tactical* — each one either declares a phase transition, repairs a reliability failure (for example the login-node guard added after one observed violation), or reshapes the illustrative direction menu; none edits a hypothesis, chooses a configuration, or interprets a result. This clean separation is precisely what licenses us to read the trajectory *between* interventions as the agent's own behaviour.

C.3 The tool-access productivity shift

The sharpest quantitative signal in the programme is the change in productivity at the Phase-1/1b boundary, where literature and code access were introduced (Table 4). Adding this access did *not* raise the magnitude of the best per-hypothesis gain — the largest gains had already occurred in Phase 1 — but it changed the *kind* of hypothesis the agent proposed: the fraction of hypotheses that modify code rather than only configuration rose from a few percent to roughly three-quarters (Figure 1). Without this transition the run would almost certainly have terminated at the Phase-1 configuration ceiling. We read this as direct evidence that the agent's *action surface*, not its reasoning capability, is the dominant lever on long-horizon productivity: the same underlying model became markedly more productive purely because it was permitted and equipped to change more of the system.

C.4 Attribution of the accuracy gain

Attributing the overall CIFAR-10 improvement of roughly twenty-seven percentage points to individual accepted hypotheses yields a strongly long-tailed distribution (Figure 2). The single largest gain comes from the very first accepted hypothesis — adding a feed-forward sublayer to every channel-attention block — which alone accounts for roughly four-fifths of the total; every subsequent accepted hypothesis is a comparatively modest refinement. This shape characterises the trajectory as a whole: a few early structural decisions dominate the outcome, while the long autonomous tail contributes steady but sharply diminishing returns. It also tempers the headline result — the agent's decisive contribution was concentrated in a handful of moves, with the remaining weeks spent refining rather than transforming the design.

Table 3: Complete record of human interventions during the research programme. All interventions are strategic; all tactical decisions (hypothesis selection, configuration editing, code writing, and result interpretation) were made by the agent.

When	Type	What changed
Phase 1 start	Research brief issued	Scientific question, design constraints, parameter and compute budget, workflow template.
During Phase 1	Reliability repair	Brief amended to instruct the agent to end its turn after submission rather than poll the scheduler itself.
Phase 1 / 1b boundary	Tool-surface expansion	Literature access enabled; brief amended to require literature grounding and to permit code-level changes.
Phase 1b / 2 boundary	Scale transition	Parameter budget raised; dataset difficulty increased; scale-transition nudge ("re-tune training recipe at the new scale").
During Phase 2	Research-direction nudge	Plateau flagged; brief amended to prioritise structural innovation over configuration tuning.
During Phase 2	Memory discipline	Full research-log read required at session start; compact agent-authored synthesis document introduced.
Phase 2 / 3 boundary	Novelty discipline	Brief amended with a "bold structural hypothesis" requirement, a mandatory per-entry novelty self-audit, and an illustrative menu of five example directions; the agent's Phase 3 set ended up including adaptations of all five.
Phase 3 start	Benchmark transition	Primary benchmark switched to ImageNet-1K, with screening and validation protocols.
End of Phase 3	Programme closure	Programme stopped after eighteen Phase-3 hypotheses, given an observed saturation around a low-rank attention champion. No new direction prescribed.

C.5 Workflow-induced versus inherent bias

Figure 3 summarises a qualitative decomposition of the agent's observed behavioural biases — greedy anchoring, post-failure risk aversion, and the overall drift toward incrementalism — into a component attributable to the greedy commit-or-discard workflow we imposed and a residual attributable to the agent's own tendencies. The workflow-induced component is the larger of the two and is directly addressable by a workflow change (maintaining a population of parallel champions, budgeting mandatory moonshots, supporting explicit backtracking); the residual is smaller but harder to modify. We stress that this is a qualitative decomposition, offered as a structured hypothesis for a future multi-workflow study to test, and not a controlled ablation — it is the reasoning behind our central claim that the loop is at least as consequential as the model.

C.6 Cross-scale reversals

A subset of Phase-1 findings did not transfer to later phases, and in several cases reversed outright (Figure 4). Recipe settings carefully tuned at the small (5.8M, CIFAR-10) scale — Mixup, label smoothing, and drop-path — were silently carried into the larger (22M, CIFAR-100) scale, where two of them flipped sign, and recovering the lost accuracy required several hypotheses of re-tuning. The agent did not automatically flag the scale transition; re-validation was triggered only by an explicit human nudge at the phase boundary. The practical lesson for autonomous-research systems is that a decision validated at one scale should be treated as *locally valid* and re-checked when the regime changes; the simplest mitigation is to annotate every accepted finding with the regime in which it was validated and to re-validate the leading findings whenever that regime changes.

Table 4: Phase-1 → Phase-1b tool-access transition. Literature access did not increase per-hypothesis gain magnitude but changed the *kind* of hypothesis proposed, breaking the config-saturation wall. Columns: Phase 1 (no literature tools) vs Phase 1b (with tools).

Metric	Phase 1	Phase 1b
Code changes (frac. of hypotheses)	2/42 = 5%	16/21 = 76%
Novel mechanisms introduced	2	8
Distinct papers cited	0	24+
Avg gain / successful hyp.	+1.70 pp	+0.19 pp
Avg gain / succ. hyp., excl. H1	+0.29 pp	+0.19 pp

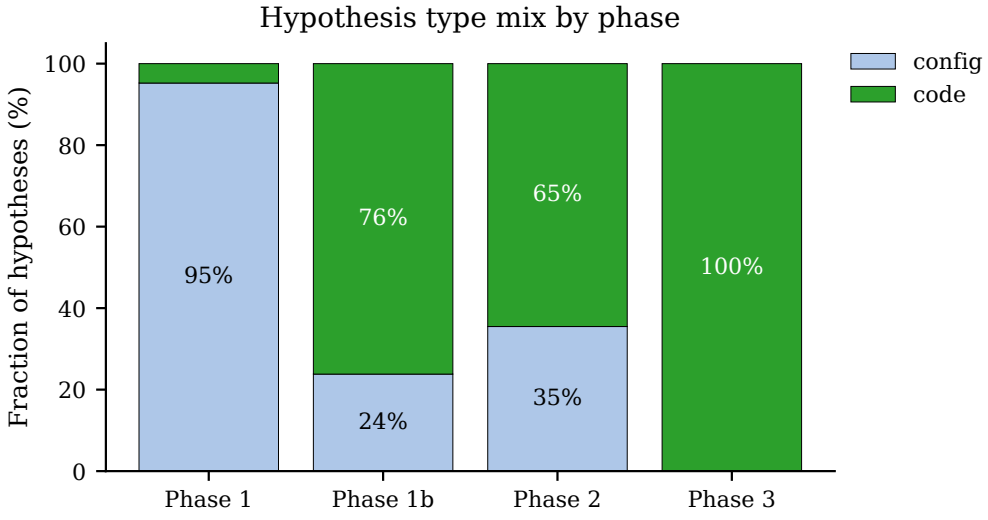


Figure 1: Mix of hypothesis types by phase. The fraction of code-modifying hypotheses (versus configuration-only tweaks) rises sharply at the Phase-1/1b literature-access boundary and remains high thereafter, quantifying the qualitative shift discussed above: the tool grant redirected the agent from tuning knobs to editing mechanisms.

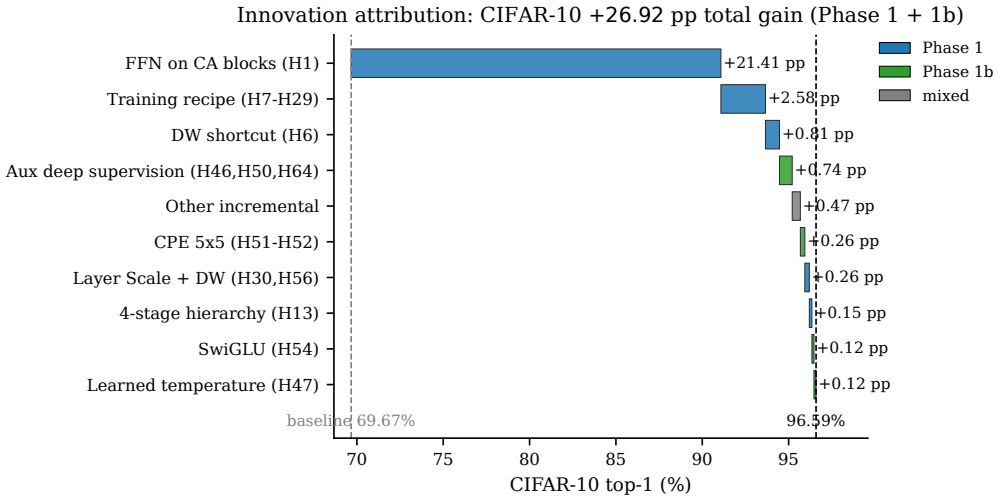


Figure 2: Innovation attribution. The total CIFAR-10 improvement (+26.9 pp) is strongly long-tailed: the first accepted hypothesis accounts for roughly four-fifths of the gain, and every later accepted hypothesis is a modest refinement.

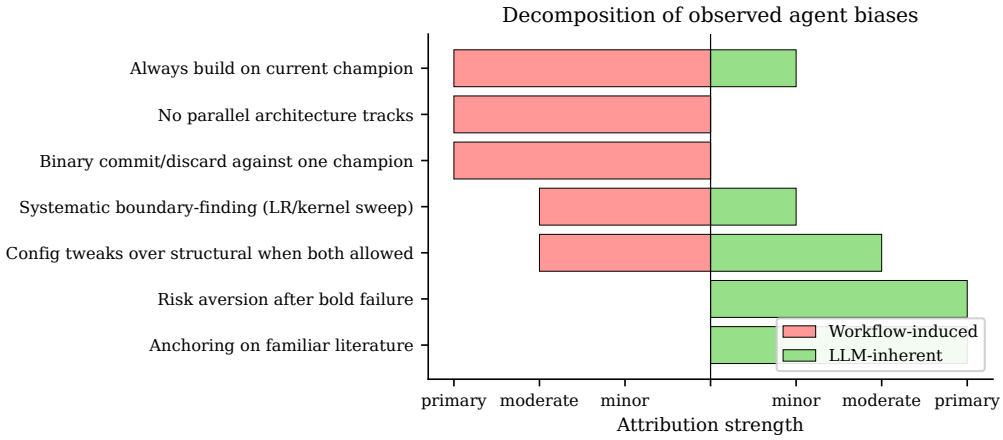


Figure 3: Diverging-bar decomposition of the observed behavioural biases. Left bars are the component attributable to the imposed greedy commit-or-discard workflow; right bars are the residual attributable to the agent's own tendencies. The workflow-induced component is directly addressable by a workflow change; the residual is smaller but harder to modify. This is a qualitative decomposition, not a controlled ablation.

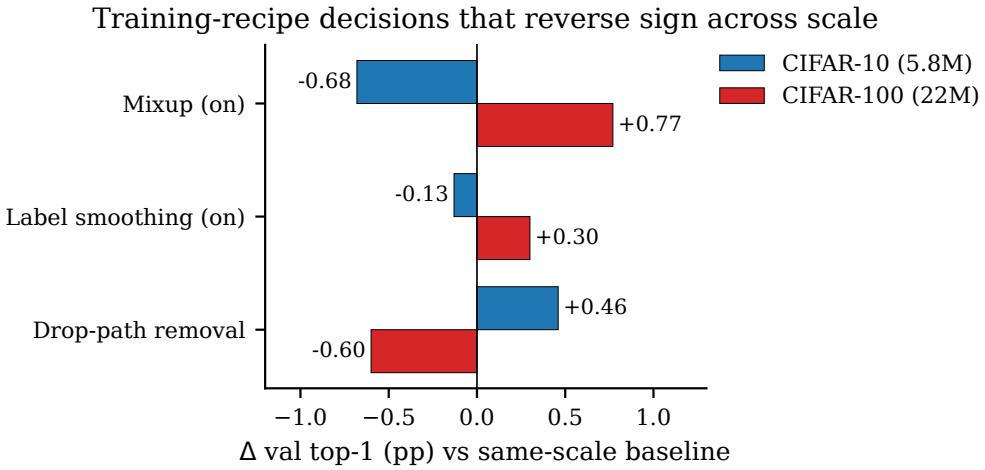


Figure 4: Cross-scale reversals. Three training-recipe decisions tuned at the small (5.8M, CIFAR-10) scale reversed sign at the larger (22M, CIFAR-100) scale: Mixup and label smoothing, mildly harmful at the small scale, became beneficial; drop-path removal, beneficial at the small scale, became harmful.